SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name

Pecel Lele

Project / Solution Name

Adept

Name (Leader)

David Kurniawan

Email & Phone (Leader)

[To be completed in the signed version]

University Affiliation(s)

Xiamen University Malaysia

Live Prototype URL

https://adept-my.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/pecel-lelele/adept

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Next.js serves a public interface in three languages and forwards requests to a private FastAPI service. Signed accounts and server-side membership checks protect each workspace. Replit's managed PostgreSQL stores profiles, reference catalogues, snapshots, programmes and progress; local development and CI use PostgreSQL with pgvector. A Python launcher checks API and database readiness and does not migrate or seed on startup.

A daily GitHub Actions schedule sends an authenticated trigger to the deployed application. Its shared worker retrieves public research, job feeds and course facts, runs bounded checks, calls the dedicated RunPod MiniLM endpoint for research embeddings and publishes validated, dated results in one transaction. Page reads reuse stored results, and historical fixtures or source outages are labelled on screen.

Deterministic matching and pathways combine confirmed skills with catalogue records and scenario estimates. Optional Gemini embeddings suggest unfamiliar skills, subject to user confirmation. Optional GLM-5.3, with Gemini failover, selects checked Adi answer IDs, and only server-written text is displayed.

The system block diagram is in the appendix.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/ attributed.

Name & Version	Purpose in Project	License
Python 3.11; Node.js 24; pnpm 10.29.3	API and pipeline runtime; web runtime and package management	PSF-2.0; MIT; MIT
Next.js 16.4.2; React and React DOM 19.3.1; TypeScript 5.9.3	Public web interface, server-side proxy and static typing	MIT; MIT; Apache-2.0
Tailwind CSS and @tailwindcss/postcss 4.1.18; class-variance-authority 0.7.1; clsx 2.1.1; tailwind-merge 3.3.1	Styling and UI component variants	MIT, except class-variance-authority Apache-2.0
next-intl 4.14.9; Recharts 2.15.4; Three.js 0.180.0; lucide-react 1.61.0	English, Malay and Chinese interface; chart components; 3D Adi mascot; icons	MIT, except lucide-react ISC
@fontsource-variable/inter 5.3.0	Installed font package; the interface currently renders with system fonts	MIT package; OFL-1.1 font
openapi-fetch 0.14.1; openapi-typescript 7.9.1	Typed API client generated from the API contract	MIT
FastAPI 0.161.1; Pydantic 2.14.1; pydantic-settings 2.16.1; Uvicorn 0.54.0	API framework, strict validation, configuration and ASGI server	MIT, except Uvicorn BSD-3-Clause
psycopg 3.3.6 and psycopg-pool 3.3.3; PostgreSQL 16 with pgvector	Relational and vector storage, connection pooling	LGPL-3.0-only; PostgreSQL License
PyMuPDF 1.28.2; python-multipart 0.0.32; HTTPX 0.28.1	CV text extraction and report export; file uploads; outbound requests	AGPL-3.0 or Artifex commercial; Apache-2.0; BSD-3-Clause
RunPod SDK 1.12.0; sentence-transformers 6.2.0; Transformers 5.20.2; PyTorch 2.14.1	Hosted MiniLM embedding worker in a separate container image	MIT; Apache-2.0; Apache-2.0; BSD-3-Clause

pytest 9.1.1; Ruff 0.17.2; ESLint 9.39.5; eslint-config-next 16.4.2; Playwright 1.64.1; Prettier 3.9.10; PyYAML 6.0.3; actionlint 1.7.7; Mermaid 11.12.0; uv	Tests, linting, formatting, browser journeys, workflow checks, diagrams and Python environments	MIT, except Playwright Apache-2.0 and uv MIT or Apache-2.0
Docker Engine and Compose	Local PostgreSQL for development and tests only	Apache-2.0 for open-source components

Versions come from the project's manifests and lock files on 1 November 2026. PyMuPDF's AGPL-3.0 terms need review, or a commercial licence, before commercial distribution.

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
arXiv API (Cornell University), cs.AI listings	Public research observations for trend signals; the October 2024 replay stays labelled historical	Free public API
UN Statistics Division SDG API	Official goal and target definitions; Adept's links are team alignments, not UN endorsement	Free public API
Microsoft Learn Catalog API; Coursera course API; course provider web pages	Course availability, duration and fees, with retrieval dates	Free public endpoints; provider terms apply
Bank Negara Malaysia Open API	Exchange rates for course prices quoted in foreign currency	Free public API
Greenhouse, Lever and Ashby public job-board APIs	Employer job feeds, filtered to Malaysian locations	Free public endpoints

Himalayas, Jobicy, Remote OK, Remotive	Dated remote job postings	Free public access
Workday and Jibe career sites	Legacy connectors in the command-line tool, present in the committed job extract; not used by automatic refresh	Free public pages
Careerjet, Jooble, Web3.career	Optional keyed job connectors, skipped when no key is configured	Keyed access; not active
OpenDOSM monthly Labour Force Survey	Malaysian national labour context, not role-level demand	Free public data
RunPod Serverless: MiniLM endpoint and GLM-5.3 endpoint	Batch research embeddings; optional Adi answer selection	Sponsor-provided credits, then paid per use
Google Gemini API: 3.5 Flash Lite, 3.6 Flash, 3.1 Flash Lite, gemini-embedding-001	Optional Adi failover and unfamiliar-skill suggestions; optional trend matcher after an equivalence test	Free or paid tier by account; off by default
GitHub REST API (workflow dispatch)	Optional administrator fallback that starts the trend-refresh workflow	Free with a scoped token

The U.S. BLS 2023-2033 projections are a committed, dated extract and are not called per user. MiniLM weights are downloaded from Hugging Face when the worker image is built, and the model runs on RunPod.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale	Public web server and private API for the live prototype	Sponsor-provided Replit Core; Asia region, 2 vCPU and 4 GiB at the proof-of-use capture

Replit managed PostgreSQL	Separate development and production application databases; signed local accounts live here	Usage-billed under the Replit account
RunPod Serverless	MiniLM embedding worker and optional GLM-5.3 endpoint; an earlier GPU Pod test was a one-off	Pay per use from US$25 sponsor credits; region not recorded
GitHub repository, Actions and Container Registry	Source code, CI, daily refresh trigger, worker image and manual fallback refresh	Public repository on GitHub-hosted runners
Docker Compose (local)	PostgreSQL with pgvector for development and tests only	Local machine, not deployment

The prototype deploys from the repository root as one public Next.js server and one private FastAPI process, started by a Python supervisor.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
all-MiniLM-L6-v2, revision 1110a243	Hugging Face weights on the team's RunPod Serverless worker	Pre-trained; 384-dimensional embeddings of public papers and catalogue skills	Apache-2.0
GLM-5.3 (Z.ai)	RunPod, for optional Adi answer selection	Pre-trained hosted inference; selects answer-block IDs only	Z.ai and RunPod terms
Gemini 3.5 Flash Lite, 3.6 Flash, 3.1 Flash Lite	Google Gemini API, optional Adi failover	Pre-trained hosted inference; off by default	Gemini API Terms

gemini-embedding-001	Google Gemini API, optional	Pre-trained; 768-dimensional skill suggestions and an optional trend matcher	Gemini API Terms

No model was fine-tuned or self-trained. By default the trend matcher uses MiniLM and moves to Gemini embeddings only after a passing method-equivalence report. The default runtime uses no language model.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
142 skills, 48 roles (33 target) and role requirements	Team curation; O*NET 31.0 (onetcenter.org)	Public-derived, curated	O*NET CC BY 4.0, with stated exceptions	N
arXiv cs.AI paper metadata and mention counts	arxiv.org	Public	Metadata CC0 1.0; papers keep own licences	Y: public author names only
U.S. occupational projections 2023-2033	bls.gov	Public, derived	U.S. public domain; not a Malaysian forecast	N
October 2024 trend replay and synthetic fallback	Team-created from dated arXiv metadata	Public-derived, synthetic	Team-created; labelled on screen	N
17 SDG goals, 169 targets; 88 role and 167 skill alignment rows	UN SDG API; team curation	Public, curated	UN terms; no UN endorsement	N
104 courses and skill prerequisites	Microsoft Learn; Coursera; named providers	Public facts, curated	Rights remain with providers	N
1,175 job postings; 14 certifications; exchange rates	Job boards above; certification issuers; BNM	Public, curated	Each source's terms	N: employers named only
Monthly Labour Force Survey; MyCOL 2024/2025	OpenDOSM; TalentCorp	Public aggregates, crosswalk	CC BY 4.0; TalentCorp terms	N

Demonstration profiles, employers, CVs and fixtures	Team-created	Synthetic	Team-created	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Anthropic Claude Code; versions varied	Code, debugging, documentation, submission drafting, pitch deck and demo-film composition	Major
OpenAI Codex; versions varied	Debugging, implementation review, code and document corrections	Major
Greptile; version not recorded	Automated pull-request review comments	Moderate
Replit Agent; version not recorded	Deployment configuration, publishing and repository exploration	Minor
Google Antigravity agent; version not recorded	Repository exploration	Minor
ElevenLabs; stock voice, no cloning	Demo-film narration only; not part of the application	Minor

Non-generative presentation tools: Remotion 4.0.538 for demo-film composition, Mixkit for media assets, and Google Fonts for typefaces.

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team set Adept's product direction and reviewed changes through pull requests before merging. Claude Code produced much of the source and document drafts, and Codex helped debug and review. Team members remain responsible for all accepted code and claims. Changes must pass automated lint, type, contract, attribution and fairness checks, and factual claims are checked against source records and dates. Model output cannot create catalogue entries or personnel decisions.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Core subscription, Autoscale deployment, managed PostgreSQL	Public prototype, private API and persistent application records; Replit Agent is declared in Section 4.3
RunPod	US$25 credits; GPU Pod test; Serverless MiniLM and GLM-5.3 endpoints	Research embeddings and optional Adi inference; the one-off Pod test validated batch encoding

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Both are important to the current deployment and both can be

replaced. Replit hosts the web server, API and database; moving would mean migrating the same schema, sessions and backups to another host. RunPod provides MiniLM encoding and optional Adi inference; replacing MiniLM needs a method-equivalence check before any scored series changes. Matching and plan reads are deterministic and make no per-page model call, and provider outages keep dated data or show an unavailable state, so the core journey continues without either model service.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State “No user data is collected” if applicable.

[Maximum 150 words]

The demonstration uses synthetic identities and CVs. Signed accounts can store names, emails, password hashes, confirmed skills, preferences, memberships, plans, progress and evidence links in PostgreSQL. CV files are parsed in memory and not retained. Members opt in to share skills and progress, and personal plans and assistant conversations stay separate. Authentication, server-side membership checks, upload limits and HTTP-only signed cookies protect this data.

Optional Gemini suggestions receive short phrases with contact details removed. Optional Adi providers receive filtered context and server-written candidate answers; redaction reduces but cannot eliminate the chance of personal information leaving the service. MiniLM receives only public research and catalogue text.

Data is kept until the user deletes the account, and account

export and deletion are built in. Automatic retention schedules and backup-erasure testing are not yet in place, and Adept makes no PDPA certification claim.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination. Adi can only select server-written answers; validation rejects unknown IDs, and a fixed guide takes over on failure. An answer can still be selected for the wrong question, so people keep the final decision.

Signal quality. arXiv attention and a U.S. labour baseline cannot measure Malaysian occupation demand. Scenarios and exposure values are heuristics, not calibrated probabilities. Adept labels them as estimates and shows source dates and fallbacks.

Bias. Scores read only skills, trend and interest. Counterfactual tests found no rank change in 75 swaps, but this does not prove universal fairness.

Coverage. The catalogue covers selected roles and courses, and only 38% of openings map to them today. Fees and openings change.

Provider dependency. Hosting and encoding rely on Replit and RunPod, with Gemini failover for Adi. Rate limits and bounded requests limit runaway use. Production refresh still needs confirmation.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☐ The prototype described here is accessible to judges through the links provided in Section 1.

☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date]

Appendix: System Block Diagram

